



HACKATHON 2026

THE AGORA
HACKATHON
2026
IS A
VIRTUAL
EVENT
DESIGNED
TO
HELP
TECH
ENTREPRENEURS
GROW
THEIR
BUSINESSES
AND
CREATE
NEW
TECHNOLOGICAL
SOLUTIONS
FOR
THE
FUTURE.

Pragati

Round: II

IDEA SUBMISSION



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POWERED BY  **KNOTIC**

Team Details:

S.No	Team Member Name	Team Member Role	Team Member Mail ID
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Track - Problem Statement:

Track Chosen: Track 1- Coordinated AI interview Panel

- One Problem Statement

Problem Statement Chosen: PS11 - Adaptive Voice Interview Platform

Focus: Real-time, multi-role interview simulation with collaborative turn-taking and interruption.

Problem Description:

Traditional mock interviews follow static, ungrounded scripts that ignore user context and lack multi-persona pushback. Job-seekers struggle to prepare for real-world panel dynamics (such as technical correctness versus business impact) and receive only generic advice rather than actionable, transcript-linked evidence, hindering confidence and readiness for modern hiring rounds.



Proposed Solution / Idea Overview:

PathFinder is an end-to-end career decision intelligence platform. It starts with a tiered, cost-optimized job ingestion engine pulling live opportunities from corporate ATS endpoints and government portals. Users see a personalized dashboard scoring their resume against real jobs. Within Apply Studio, candidates tailor applications and track statuses.

Crucially, candidates then launch a live, adaptive AI panel interview grounded in the specific job description. Integrating Agora WebRTC for media and Gemini Live WebSockets for ultra-low-latency reasoning, the system simulates a real 3-member panel (Technical, Product, Hiring Manager). A deterministic server-side moderator uses client-side VAD to manage turn-taking and cross-examination—if a candidate gives a technically correct answer but misses business impact, the Product Manager interrupts to challenge them. The session ends with an evidence-linked assessment report, mapping verbatim transcript citations to actionable learning pathways.

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Full Solution Architecture & Lifecycle

- **• STAGE 1: INGESTION & DISCOVERY**
 - - Autonomous crawler fingerprints domains and queries public JSON boards (Greenhouse, Lever, Zoho, Keka, Ashby).
 - - Jobs are parsed via Gemini into our taxonomy, classified into Government vs. Private, and deduplicated.
- **• STAGE 2: APPLY & TRACK**
 - - Match % scores resume skills against Job description. Candidates mark status (Applied/Not Applied) with timestamp tracking.
- **• STAGE 3: ADAPTIVE INTERVIEW PANEL (PS11)**
 - - Agora carries candidate video/audio, and Gemini Live WebSocket manages real-time reasoning.
 - - Server-side Moderator uses browser VAD to manage multi-role turn-taking. If candidate correctness is high but impact is omitted, Product persona challenges.
- **• STAGE 4: EVIDENCE REPORT**
 - - Score trend dashboard logs progress, and assessors generate transcript-linked evidence lines.





Technology Stack & Cloud Scaling

- **CORE BACKEND**
 - - FastAPI (Python 3.9) with SQLAlchemy ORM (Cloud SQL Postgres in production).
 - - Stdlib-only cryptography (PBKDF2 + JWT) ensures maximal security compliance.
- **AI & MEDIA CONNECTIVITY**
 - - google-genai SDK for low-latency bidirectional WebSocket audio streaming.
 - - Agora Web SDK NG for candidate camera/mic and AI bot audio injection browser-side.
- **CLOUD DEPLOYMENT & GOVERNOR**
 - - Single lightweight Docker container deployed on Google Cloud Run.
 - - Session affinity maintains active WebSocket connections mapped to warm instances.
 - - Cron purge deletes raw turns past 180 days after copying cited text inline.



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Agora RTC & Gemini Live Synergy

● AGORA WEBRTC MEDIA TRANSPORT

- • Browser-Side Audio Injection:
 - AI audio is published straight from the browser into the Agora channel (UID 90000). Bypasses server WebRTC overhead.
- • Simulated Live Video Tiles:
 - Agora carries candidate video beside responsive interviewer avatars that pulse in size during turn speaking states.
- • Native Network Resilience:
 - Ensures high-fidelity connection, automatic reconnection and audio worklet synchronization.

● GEMINI LIVE ASYNC AUDIO BRAIN

- • Native Bidirectional WebSocket:
 - Bypasses slow STT-Text-TTS loops. Audio-in to Audio-out minimizes processing latency.
- • Client-Side VAD (Turn Control):
 - Hysteresis algorithm running in AudioWorklet isolates genuine candidate speaking starts/ends, filtering environmental clicks.
- • Symmetric Barge-In:
 - The instant the applicant starts talking, VAD flushes local playing audio buffer instantly on the frontend.

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Cost-Optimized Job Ingestion Pipeline

- **TIER 1: ATS JSON ENDPOINTS (FREE)**
 - - Direct HTTP REST requests to public ATS boards (Greenhouse, Lever, Zoho, Keka, Ashby). Bypasses metered rendering, keeping 85% of crawls free.
- **TIER 2: DISCOVERY CRAWLER (FREE)**
 - - Crawler fingerprints domains. Standardized redirects (careers.domain.com -> greenhouse.io/domain) auto-extract and cache tokens.
- **TIER 3: BRIGHT DATA WEB UNLOCKER (METERED)**
 - - Used strictly for custom-portal holdouts (TCS, Infosys) and Government recruiter portals (UPSC, state PSCs) that require browser rendering.
- **COST GOVERNOR**
 - - Capped requests per domain-day, per-run request thresholding, and live billing cost log (est_cost_usd ledger) prevent runaway bills.





Business Value & Hackathon Impact

- • **SCALABLE MOCK INTERVIEWS**
 - - Access to hyper-realistic mock boards (Tech, Product, HR) with unique voices, cooperating via a shared context.
 - - Closed-loop learning: recommended courses link directly to the report to remediate weak skills.
- • **EVIDENCE-LINKED ASSESSMENTS**
 - - Every score and critique links directly to the highlighted transcript turn, giving candidates clear auditability.
- • **MARKET ALIGNMENT**
 - - Dual sectoring of Private and official Government PSU jobs prepares candidates against real-world localized parameters.

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Additional Notes: (If Any)

CORE SYSTEM HIGHLIGHTS:

- Cost-Minimized Job Ingestion: Crawls 12,000 domains for free by fingerprinting Greenhouse, Lever, Workday, Zoho, and Keka public APIs. Utilizes Web Unlocker strictly as a fallback.
- Client-Side VAD (Hysteresis): Energy-based detection in the browser's AudioWorklet manages barge-in & turn boundaries, passing manual activity triggers to the server's Moderator.
- Non-Scrapable Score Integrity: Python analyst computes overall scores arithmetically to bypass adversarial prompt-injection (e.g. rating overrides in resumes).
- Order Invariant Purge: background cron drops PII transcripts after 180 days, copying cited quotes inline to maintain clickable evidence citation validity.